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REVIEW ARTICLE

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Chemical restraint: A qualitative synthesis review of adult service user and staff experiences in mental health settings

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Abstract

With an imperative to reduce or eliminate the use of coercive practices in mental health care it is important to understand the experience of service users and staff. This review aimed to synthesize qualitative studies, published between 1996 and 2020, reporting on mental health service users' and staff's experiences of chemical restraint. The databases PsycINFO, CINAHL, MEDLINE, Embase, Emcare, Web of Science, and Scopus were searched. Three analytic themes were identified from 17 included articles, synthesizing the experiences of service users and staff. These were "Unjustified versusjustified," "Violence versus necessity," and "Reflecting back: Positives and negatives." Service users viewed chemical restraint as an unjustified response to "behaviors of concern" and experienced it as a violent act with negative outcomes, although some saw it as necessary in retrospect and preferred it to other forms of coercion. Staff generally viewed it as a justified response to "behaviors of concern" and experienced it as appropriate within the constraints of staff numbers and limited alternatives. These findings identify nuances not apparent in the literature, which has generally conflated all forms of coercive practices.

KEYWORDS
chemical restraint, coercive practices, forced medication, involuntary medication, psychiatric, qualitative synthesis

1 | INTRODUCTION

The use of coercive practices (including seclusion and physical, mechanical, and chemical restraint) is commonplace in the "management" of "behaviors of concern" in mental health settings, with these practices often being used in combination (Gerace & Muir-Cochrane, 2019). Behaviors of concern include extreme agitation, violence, aggression, and self-harm. It is recognized that these behaviors result from the complex interplay between the service user and the physical, social, and political environment of mental health settings (Brophy et al., 2016; Gerace & Muir-Cochrane, 2019; Holmes et al., 2013; McKeown et al., 2020).

There is an international imperative to reduce or eliminate the use of coercive practices, including chemical restraint, in mental health care. Chemical restraint (the use of medication as a form of restraint) occurs in various settings, including prehospital situations such as during ambulance call outs, as well as in emergency departments, acute psychiatric inpatient units, or psychiatric intensive care units (Isbister, 2017; Taylor et al., 2017; Wilhelm et al., 2008; Zun, 2018; Zun et al., 2017). While there has been sustained focus on other forms of restraint, chemical restraint has received less attention. This study provides new understandings for mental health nursing by synthesizing the international qualitative literature on chemical restraint in mental health settings.

2 | BACKGROUND

There is definitional opaqueness in regard to the term chemical restraint, which is often used interchangeably with terms such as rapid tranquillization, forced sedation, and forced medication (Muir-Cochrane, 2020). It is generally understood to be the "assertive

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administration of emergency medication to adults with aggressive, agitated or violent behaviours with the purpose of quick calming or sedation, diminution of symptoms, and/or to decrease the likelihood of harm to self or others" (Battaglia et al., 2003, p. 192). Chemical restraint involves the use of two different forms of medication usually administered conjointly. The first is neuroleptics for the treatment of symptoms of mental illness, and the second is anxiolytics to induce calm, generally given in an emergency when a person's behavior is considered to be out of control and the person is assessed as being a harm to self or others.

Much of our understanding of chemical restraint comes from quantitative research exploring the prevalence, risk factors, effectiveness, and medication types used in chemical restraint. In a recent project covering 22 years of research into chemical restraint, the authors conducted a systematic review of quantitative research (PROSPERO registration CRD 42017055258). A total of 311 primary studies were identified, conducted in acute psychiatric wards, general psychiatric wards, and general hospital emergency departments (EDs). A meta-synthesis of 48 retrospective audits found prevalence of chemical restraint to be 7.4% of service users admitted to these settings (Muir-Cochrane, Grimmer, et al., 2020). An analysis of 21 randomized controlled trials found that a number of drugs were used in chemical restraint, yet there was a lack of clarity in the superiority of any chemical method of restraint (Muir-Cochrane, Oster, et al., 2020). Across the body of literature, we identified tensions between using chemical restraint without producing adverse events, decision-making about when this form of restraint was needed, and finding a balance between respecting dignity while ensuring the safety of staff and service users (Muir-Cochrane et al., 2020a). Lack of homogeneity in the samples across studies was also an issue (Muir-Cochrane et al., 2020b).

An important consideration when examining coercive practices such as chemical restraint is the experience of service users and staff explored through qualitative research. Understanding these experiences would provide important evidence to inform the care of service users and provide support for nursing and other staff. Previous qualitative research on coercive practices reports overwhelmingly negative experiences of service users and staff (Cusack et al., 2016; Kinner et al., 2017), while being justified by staff to maintain safety for service users, staff, and visitors (Brophy et al., 2016; Gerace & Muir-Cochrane, 2019; Kinner et al., 2017). Research furthermore reports nurses feel fear and blame associated with restraint (Bigwood & Crowe, 2008; Muir-Cochrane et al., 2018).

There is a tendency for qualitative research of the experience of coercive practices to "bundle up" these practices into one phenomenon rather than exploring these practices individually. For example, Hawsawi et al. (2020) conducted a qualitative literature review of nurses' and service users' shared experiences of seclusion and restraint. Shared experiences included disruption in care and in the therapeutic relationship and shared negative effects. From nurses' perspectives, the experience encompassed an absence of less coercive alternatives while service users reported feeling overpowered, humiliated, and punished.

The "bundling" of coercive practices in the research literature is likely because in order for seclusion to occur physical restraint may be involved, thus exploring an experience of seclusion is also usually an experience of some other kind of restraint. Similarly, chemical restraint is often experienced in the context of physical and/or mechanical restraint. Yet chemical restraint is arguably a qualitatively different experience to physical restraint, mechanical restraint, or seclusion, and it is therefore important to explore these experiences separately. While previous reviews of qualitative studies of seclusion have been reported (Askew et al., 2019), no previous systematic reviews focusing on the experiences of chemical restraint were identified. To that end, we conducted a systematic review and thematic synthesis of qualitative studies to specifically explore the experiences of service users and the range of staff involved in chemical restraint. Given the continuing use of chemical restraint, it is important to understand the phenomenon of this practice in order to provide new understanding and potential implications for future care.

3 | AIM

This qualitative synthesis review aimed to synthesize qualitative studies, published between 1996 and 2020, reporting on mental health service users' and staff's experiences of chemical restraint in mental health settings.

4 | METHODS

This study was conducted by two researchers with an interest in exploring the practice and experiences of coercion in mental health care. The first author is a Credentialed Mental Health Nurse and senior academic. The second author is a researcher with no experience in mental health settings.

4.1 | Literature search

The databases PsycINFO, CINAHL, MEDLINE, Embase, Emcare, Web of Science, and Scopus were searched for studies published from January 1, 1996 to March 16, 2020 to ensure we captured literature reflecting contemporary mental health practices. We consulted with a university liaison librarian to determine the index and MeSH search terms. These included chemical restraint, forced medication, coerced medication, intramuscular medication, rapid tranquilization, restraint, mental health, psychiatric, emergency department, and variants of these, with further terms included that were identified in a preliminary review of the literature (see Appendix S1 in the Supporting Information).

4.2 | Data evaluation

4.2.1 | Inclusion and exclusion criteria

To be eligible for inclusion the articles had to report on qualitative peer-reviewed research, be published in English, and capture adult

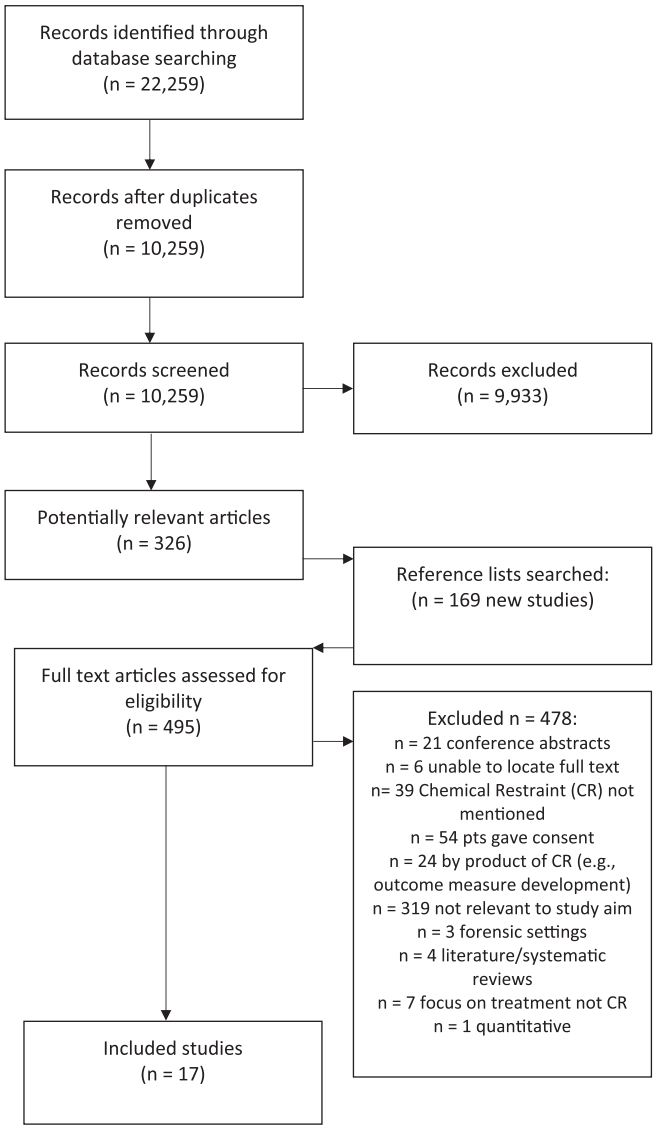

FIGURE 1 PRISMA flow chart of the search and assessment

service user and/or staff experiences of chemical restraint. Settings included inpatient and outpatient mental health settings and emergency departments (EDs). As stated earlier, we defined chemical restraint as the "assertive administration of emergency medication to adults with aggressive, agitated or violent behaviours with the purpose of quick calming or sedation, diminution of symptoms, and/or to decrease the likelihood of harm to self or others" (Battaglia et al., 2003, p. 192). Through the assessment process we encountered significant difficulties in determining whether or not articles were exploring chemical restraint. This was particularly the case with regard to articles using the term "forced medication." This term was used to describe forced medication in the context of emergency management of "behaviors of concern," forced medication for treatment purposes, with some articles including experiences of forced medication in the context of both the management of "behaviors of concern" and treatment. Only those articles explicitly referencing forced medication in the context of "behaviors of concern" were included, to ensure accordance with the definition of chemical restraint we used as behavioral management and not treatment. Two independent reviewers screened the search results for eligibility. Reference lists of included articles were screened for any articles not identified in the literature search.

4.2.2 | Critical appraisal

Included articles were critically appraised by two independent reviewers using the Qualitative Critical Appraisal Skills Programme (CASP) tool (Singh, 2013), with any disagreements resolved through discussion. The CASP checklist includes 10 questions relating to validity, study results, and applicability of results, answered "Yes," "No," or "Cannot Tell."

4.3 | Data analysis

Results of the included qualitative studies were analyzed using Thomas and Harden's (2008) Thematic Synthesis. Given the tendency to conflate multiple forms of restraint when exploring experiences of service users and staff, we began by extracting data explicitly related to chemical restraint (as per Askew et al., 2019, in their synthesis of data related to seclusion) and copying and pasting this into a text document. At times it was unclear whether the data related to chemical or other restraints. In these situations, we opted for an inclusive approach to ensure a sufficiency of data for subsequent analysis.

Data extracted from the articles was in the form of information presented in the results sections of the included articles, including participant quotes where available. Following data extraction, we synthesized the data by progressing through three stages. First, we coded the data line by line. Descriptive themes were then developed, followed by inductive thematic analysis. The authors independently coded the data and then met to discuss the development of descriptive and inductive themes.

5 | RESULTS

5.1 | Summary of included studies

The PRISMA flow chart (Figure 1) shows the results of the searching and assessment processes. Following these processes, 17 articles were included in the thematic synthesis. The majority of the articles discussed the experience of chemical restraint in the context of other forms of restraint, with only three articles focusing on chemical restraint (Mayers et al., 2010; Vuckovich & Artinian, 2005; Yap et al., 2017).

The articles are summarized in Table 1. A number of terms were used to describe the phenomenon, mainly forced medication (mentioned in 7 articles) and chemical restraint (mentioned in 6 articles), as well as sedation (n = 3 articles), pharmacological restraint (n = 2), rapid

TABLE 1 Summary of articles included in the review

First author, date	Country	Population	Setting	Aim	Chemical restraint nomenclature	Relationship between researcher(s) and participants	Data collection	Method of analysis
Chambers et al., 2014	UK	Nineteen detained service users	Psychiatric inpatient	To report on the experiences of detained mental health services users	Chemical restraint	Participants were interviewed by either a service user researcher or a clinical psychologist, all of which were members of the research team	Semi-structured interviews	Thematic analysis
Duxbury, 1999	UK	Thirty-four mental health nurses & 32 general nurses	Psychiatric inpatient & acute general setting	To explore similarities and differences in nurses' experience of service user aggression	Chemical restraint; Sedation	The relationship is unclear	Critical incident technique	Content analysis
Gallop et al., 1999	Canada	Ten inpatient women who have a history of childhood sexual abuse	Psychiatric inpatient	To explore the experiences of women who were hospitalized in psychiatric settings, restrained, and given forced medication	Forced medication	The relationship is unclear	Semi-structured interviews	Content analysis
Johnston & Kilty, 2016	Canada	Eight security guards	Psychiatric inpatient	To explore (1) How do private security guards neutralize and perceive their role in perpetrating violence by using force and punishment against "unruly" psychiatric service users?; (2) How do guards feel about their status in relation to other key actors in hospital settings such as nurses, service users and other guards?	Chemical restraint	The research is a former security guard and participants were the researcher's former colleagues	Semi-structured interviews	Not stated
Katsakou et al., 2012	UK	Fifty-nine involuntary service users	Psychiatric inpatient	To explore involuntary service users' retrospective views on why their hospitalization was right or wrong	Forced medication	The research team included service users (involved in the design, data collection, and analysis) as well as those with a background in psychiatry, psychology, sociology and nursing	Semi-structured interviews	Grounded Theory and thematic analysis

TABLE 1 (Continued)

First author, date	Country	Population	Setting	Aim	Chemical restraint nomenclature	Relationship between researcher(s) and participants	Data collection	Method of analysis
Kuosmanen et al., 2007	Finland	Fifty-one service users	Psychiatric inpatient	To find out whether service user had experienced deprivation of their liberty during psychiatric hospitalization and to explore their views about it	Forced medication	The data were collected by four psychiatric nurses who were not working in the recruitment setting	Semi-structured interviews	Inductive content analysis
Ling et al., 2015	Canada	Fifty-five service users	Psychiatric inpatient	To examine debriefing data to understand experiences before, during, and after a restraint (seclusion, chemical, and physical) event from the perspective of service users	Chemical restraint	Qualitative data was obtained from a Restraint Event Client Debriefing and Comments Form, completed by staff in the setting	Debriefing data	Thematic analysis
Looi et al., 2015	Sweden	Nineteen service users	Psychiatric inpatient	To describe how people who self-harm perceive alternatives to coercive measures in relation to actual experiences of psychiatric care	Coerced medication	No relationship between the researchers and participants (data collected via an anonymous self-report)	Written self-report	Qualitative content analysis
Mayers et al., 2010	South Africa	Sixteen service users	Psychiatric inpatient	To explore the perceptions and experiences of service users who have been exposed to sedation, seclusion, and restraint	Sedation; Chemical restraint; Pharmacological restraint	The research team comprised service users, service providers, and academic researchers	Focus group	Thematic analysis
Olofsson et al., 1999	Sweden	Ten physicians	Psychiatric inpatient	To explore physicians' experiences with using coercion	Chemical restraint; Forced injection	No relationship between the researchers (academics) and participants	Semi-structured interviews	Thematic content analysis
Price et al., 2018	UK	Twenty ward-based clinical staff	Psychiatric inpatient	To explore staff perspectives on factors influencing the success or failure of de-escalation techniques for the management of violence and aggression in mental health settings	Forced medication	The interviewer was a registered mental health nurse	Semi-structured interviews	Framework analysis

(Continues)

TABLE 1 (Continued)

First author, date	Country	Population	Setting	Aim	Chemical restraint nomenclature	Relationship between researcher(s) and participants	Data collection	Method of analysis
Rose et al., 2015	UK	Thirty-seven service users; 48 nurses	Psychiatric inpatient	Experiences and perceptions of life in acute mental health settings	Forced medication; Rapid tranquillization	The service user focus groups were facilitated by service user researchers; the nurse focus groups were facilitated by nurse researchers	Focus group	Thematic analysis
Salzmann-Erikson et al., 2008	Sweden	Eighteen nurses	Psychiatric intensive care units (PICUs)	To describe the core characteristics of a PICU in Sweden and to describe the care activities provided for service users admitted to the PICUs	Rapid tranquillization; Sedative medications	The relationship is unclear	Critical incident technique	Content analysis
Sibitz et al., 2011	Austria	Fifteen involuntary service users	Psychiatric inpatient	To establish a typology of coercion perspectives and styles of integration into life stories	Forced medication	It is stated that the interviews were undertaken by a researcher with previous relationship to the participants	Semi-structured interviews	Thematic analysis
Vuckovich & Artinian, 2005	USA	Seventeen psychiatric nurses	Psychiatric inpatient	To explore psychiatric nurses' experiences of administering medication to involuntary psychiatric service users	Forced medication	The relationship is unclear	Semi-structured interviews	Grounded Theory
Wynn, 2004	Norway	Twelve psychiatric inpatients	Psychiatric inpatient	To explore service users' experiences regarding restraint	Pharmacological restraint	The relationship is unclear	Semi-structured interviews	Grounded Theory
Yap et al., 2017	Australia	Thirteen service users	Emergency Department	To explore service users' perceptions and experiences of sedation during behavioral emergencies	Sedation	It is stated that the interviewer was not involved in the clinical care of participants	Semi-structured interviews	Thematic analysis

tranquillization (*n* = 2), coerced medication (*n* = 1), forced injection (*n* = 1), and sedative medications (*n* = 1). The majority (*n* = 10) focused on the experience of service users followed by nurses (*n* = 3), with the other articles focusing on the experience of physicians (*n* = 1), ward-based clinical staff (*n* = 1), both service users and nurses (*n* = 1), and one article focused on the experience of security guards. Settings were mainly psychiatric inpatient units (*n* = 15) with one study conducted in the ED and one in a psychiatric intensive care unit.

Results of the critical appraisal are presented in Table 2. The studies were generally well-conducted. One area where there was consistent lack of clarity in the presentation of results related to Question 6, "Has the relationship between researcher and participants been adequately considered?" Given that there are clear imbalances of power between staff and service users in the use of chemical restraint, and the potential for research participants to be less open in their responses to questions posed by researchers who are closely aligned to (or part of) the system in which chemical restraint is sanctioned, Table 1 includes a description of the relationship between researcher(s) and participant(s) for each included study. This information was either explicitly stated in the articles or extrapolated from the information provided.

5.2 | Thematic synthesis

Three analytic themes were identified as synthesizing the experiences of service users and staff. These were "Unjustified versus justified," "Violence versus necessity," and "Reflecting back: Positives and negatives." The experiences of service users and staff in relation to these themes are discussed below. Chemical restraint was generally experienced with other forms of coercion, particularly physical restraint and sometimes mechanical restraint. Hence the data presented below often include discussions of chemical restraint in the context of other forms of restraint. This is discussed further in regard to this study's limitations. Note that while authors variously use the terms "patient," "consumer," or "service user," we have used the term "service user" throughout except when quoting directly from sources that use a different term.

5.3 | Unjustified versus justified

This theme relates to how service users and staff viewed the antecedent behaviors and the response (chemical restraint). Antecedents to chemical restraint were "behaviors of concern," particularly violence and aggression. Service users generally reported these behaviors as an understandable response to their situation or the environment resulting from an escalation in the situation leading to chemical restraint. For example:

> One of the main reasons given by patient participants for behaviour that might elicit restraint or forced medication was that users were cooped up in the ward and not allowed to go outside and get fresh air. (Rose et al., 2015, p. 93)

> "I was feeling really bad being in the hospital. … she was a patient from another hospital . . . started blasting me how she didn't understand how women who had children could be suicidal. I just lost it. … I threw my tray . . . went to my room and right away they called the doctor…" (service user quoted in Gallop et al., 1999, p. 407)

In contrast, staff viewed service user "behaviors of concern" as sudden and unreasonable, such as Salzmann-Erikson et al.'s (2008) discussion of "The dramatic admission" and "Escalating behavior."

These perspectives of antecedents to chemical restraint informed how service users and staff viewed the response, that is, chemical restraint. From a service user perspective, chemical restraint was viewed as unjustified and at times as punishment and a mechanism for keeping them quiet. For example:

> Sedation was perceived more as a mechanism for keeping service users quiet, and less for therapeutic reasons. In particular, haloperidol was feared due to the side effects experienced. This medication has been given a colloquial name by those who have had this prescribed for them and is known as the "Hou jou bek pil" [keep your mouth shut pill]: "You know that tablet is like a *sjambok* [whip], it punishes you and the side effects are very severe." (Mayers et al., 2010, p. 68)

Service users in Looi et al.'s (2015, p. 98) study furthermore described feeling "as though the staff wanted to provoke frustration to justify the use of coercive measures." A contrasting perspective was provided in Yap et al.'s (2017) exploration of sedation during behavioral emergencies in the ED context. Service users here viewed chemical restraint as an appropriate response to "behaviors of concern" in this environment.

Staff reported chemical restraint as a justified response to service users' sudden and unreasonable behavior. A key justification for the use of chemical restraint was one of safety, that is, maintaining the safety of service users, visitors, and staff by quickly suppressing violence and aggression. Nurses in Rose et al.'s (2015) study saw chemical restraint as "a legitimate response to violence" (p. 93) and furthermore saw service user violence/aggression as at times being a deliberate act on the part of service users and "a tool to express their resentment at their situation on the ward" (p. 94). Vuckovich and Artinian (2005, p. 370) explained: "The process of justifying coercion allows a nurse to engage in behavior generally disapproved of while retaining a self-image of a 'good' nurse." In the studies of physicians (Olofsson et al., 1999) and security guards (Johnston & Kilty, 2016), service user behavior was also seen as a justification for chemical restraint.

TABLE 2 Critical appraisal of the reviewed articles

First author, date	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
Chambers et al., 2014	Y*	Y	Y	Y	Y	Y	Y	Y	Y	Y
Duxbury, 1999	Y	Y	Y	Y	Y	?	Y	Y	Y	Y
Gallop et al., 1999	Y	Y	Y	Y	Y	?	Y	Y	Y	Y
Johnston & Kilty, 2016	Y	Y	Y	Y	Y	?	Y	Y	Y	Y
Katsakou et al., 2012	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Kuosmanen et al., 2007	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Ling et al., 2015	Y	Y	Y	Y	Y	N	?	Y	Y	Y
Looi et al., 2015	Y	Y	Y	Y	Y	N	Y	Y	Y	Y
Mayers et al., 2010	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Olofsson et al., 1999	Y	Y	Y	Y	Y	?	Y	Y	Y	Y
Price et al., 2018	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Rose et al., 2015	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Salzmann-Erikson et al., 2008	Y	Y	Y	Y	Y	?	Y	Y	Y	Y
Sibitz et al., 2011	Y	Y	Y	Y	Y	?	Y	Y	Y	Y
Vuckovich & Artinian, 2005	N	Y	?	?	Y	?	?	Y	Y	Y
Wynn, 2004	Y	Y	Y	Y	Y	?	Y	Y	Y	Y
Yap et al., 2017	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

*Y = Yes; N = No; ? = Cannot Tell.

5.4 | Violence versus necessity

A further contrast between the views of service users and staff relates to the experience of chemical restraint itself. The experience of chemical restraint for service users was that it was a violent act leading to feelings of fear, humiliation, and powerlessness. This is particularly in the context of being held down, having clothing "ripped off" (Gallop et al., 1999, p. 411), and being injected by men. Service user quotes reported in the included studies demonstrate the extent to which chemical restraint was experienced as a violent and fearful act by service users, including:

> "jump on you with a needle"; "pound on you with the needle" (Rose et al., 2015, p. 94)

> "helpless and vulnerable"; "felt like I was being raped"; "These four attendants or nurses . . . all male . . . about six feet tall but I just kicked and screamed . . . and I kicked them in the shins . . . and they threw me down on the bed, on my face, and they restrained me and they injected me." (Gallop et al., 1999, pp. 407; 410)

> "I thought I was going to die" (Katsakou et al., 2012, p. 1173)

> "three men come and they stuck a needle in my ass" (Kuosmanen et al., 2007, p. 602)

> "I had only one thought in my head … and that was to defend myself against them … and they were five men" (Wynn, 2004, p. 131)

Service users viewed chemical restraint as "an unnecessary exercise of power" (Kuosmanen et al., 2007, p. 602) and described the need for other responses, in particular early intervention through engagement and de-escalation strategies. Service users in Yap et al.'s study (2017, p. 961), who saw chemical restraint as a justified response, also described experiencing fear—"But I feel scared 'cause I don't know why I had to be put to sleep. How did I end up here?"—and highlighted the need for information and debriefing following chemical restraint.

Despite recognition that chemical restraint was not an ideal response, and should only be used as a last resort, staff overwhelmingly experienced a lack of choice with regard to chemical restraint. For example, Olofsson et al. (1999, p. 204) reported physicians using words such as "being forced" and "not having any choice," given the need to maintain the safety of service users and staff.

Managing "behaviors of concern" was furthermore described as "a nursing responsibility" (Duxbury, 1999, p. 112). Nurses described chemical restraint and other forms of coercion as resulting from a "lack of non-pharmacological skill in managing aggressive behaviour; fear of contagion of aggression to other service users, and an inability to tolerate lengthy periods of aggression and uncertainty through inadequate regulation of fear responses," in addition to poorly resourced environments (Price et al., 2018, p. 204). In the studies of physicians and security guards, lack of choice was expressed in terms of needing to meet the needs and expectations of nurses. For example, physicians in Olofsson et al. (1999, p. 205) described: "It was one of those situations in which I felt strong pressure from the nursing staff to medicate" and "I felt rather pressed by the nursing staff." A security guard in Johnston and Kilty's (2016), p. 189) study stated: "Some nurses are like 'No, no we don't even want you to talk to him we want to hold him down and medicate.'"

5.5 | Reflecting back: Positives and negatives

The final theme, reflecting back, relates to views of the outcomes of chemical restraint. Overall, service users described negative outcomes of chemical restraint, with this practice leading to exacerbation of trauma, flashbacks, negative side effects, and loss of trust. Gallop et al. (1999, p. 411) reported chemical restraint "did not make any participant feel safe, and none found it helpful."

However, some service users did identify positive aspects of chemical restraint. For example, for service users in Rose et al. (2015, p. 94), "coercion was sometimes perceived as an appropriate staff response to other violent service users." Furthermore, when reflecting back on different types of coercive practices, chemical restraint was described by service users in Mayers et al. (2010, p. 68) as "the containment option causing the least distress, and was also perceived to best respect the service users' human rights." Wynn (2004, p. 132) stated: "Seven of the patients had received pharmacological restraint while subjected to physical restraint, but gave the impression that this intervention was overshadowed by the use of physical restraint."

Nurses and other staff similarly viewed chemical restraint as a negative experience for service users, yet in reflecting back highlighted positive outcomes. For example, nurses in Salzmann-Erikson et al. (2008, pp. 101–102) described the outcome of restraint as successful treatment and discharge:

> "The patient is admitted to ward 77 [the PICU]. The patient becomes violent and is physically restrained at the ward and given an injection [. . .]. When the patient is no longer violent, the patient can be a transferred to another ward or discharged."

Similarly, Johnston and Kilty (2016, p. 177) noted security guards "claimed that these practices benefit the patients more than it hurts them."

6 | DISCUSSION

These findings extend understandings of the experiences of service users and staff specifically regarding the use of chemical restraint with people with a mental illness. It is important to separate out the act of chemical restraint from other forms of restraint and associated service user experiences in order to understand individual restraint phenomena. Such understandings can inform future practice and build a robust evidence base to facilitate high quality care in a least restrictive environment.

The dichotomy of findings between service users and staff, whereby the majority of service users saw chemical restraint as unnecessary and unjustified with staff viewing it in the opposite sense, is a salient reminder of the seriousness of the event amidst the body of evidence that service users find the total experience of hospitalization as traumatizing (Robins et al., 2005) and chemical restraint, dehumanizing. This is recognized as sanctuary harm, the negative experience of care by service users in mental health services when in mental health distress. The negative effect of the environment on the incidence of aggression also contributed to the use of chemical restraint and this is supported by other research on restrictive measures (Brophy et al., 2016; Gerace & Muir-Cochrane, 2019).

The experience of service users finding the chemical restraint event as sudden and unreasonable could perhaps be explained by the service users' illness and escalating agitation or aggression, prompting staff to intervene swiftly in the interests of safety for all concerned. Alternatively, such reporting of experiences is likely to mean that the rapid escalation of restraint was deemed inappropriate by service users and viewed as an act of violence. Indeed, there exists a body of work on institutional violence in psychiatric settings and the differences between service users' and staff's perspectives (Holmes et al., 2013), arguing in some cases that violence is bred into mental health practices through risk culture and the normalization of restrictive practices such as chemical restraint as vital for service user and staff safety.

Our findings on chemical restraint also strongly concur with other research that has explored the experiences of nurses and service users regarding other forms of coercive practice, where both nurses and service users see coercion as negative but it is justified by nurses due to the imperative to maintain safety for all concerned (Brophy et al., 2016; Gerace & Muir-Cochrane, 2019; Kinner et al., 2017). The finding of chemical restraint being viewed as a nursing responsibility resonates with previous work indicating nurses experience fear and blame in relation to coercive practices in mental health care (Muir-Cochrane et al., 2018). Our finding regarding chemical restraint as "unjustified versus justified" is supported by other research exploring ward culture in the context of restraint minimization in health care (McKeown et al., 2020). This work identified legitimation as a "crucial discursive practice in the context of staff reliance upon coercion" (McKeown et al., 2020, p. 449), detailing the environment and lack of meaningful activities negatively affecting service user care. These authors suggest the need for the adoption of trauma informed care as an alternative supportive legitimacy.

Some service users perceived chemical restraint as causing the least distress of all forms of coercion, presumably because once medication was administered, they experienced calm and sedation. Some also acknowledged that it was necessary and appropriate, particularly in the ED context and when applied to other service users. This aspect of the findings adds to the minimal literature on chemical restraint and suggests that involving service users in decisions about their own plan of care when acutely unwell, regarding their preferred choice of restraint if and when necessary, could provide at least some autonomy to the individual in care. However, a recent systematic review on seclusion and restraint suggested that seclusion appeared to be more acceptable in comparison to forced medication (Chieze et al., 2019) suggesting a need for further examination of the acceptability of differing restrictive measures.

Our previous review on the quantitative research on chemical restraint showed the lack of heterogeneity in studies (in terms of the diversity of methods and settings) limiting the drawing of clear

conclusions (Muir-Cochrane et al., 2020b); so too is the case for studies on the most effective interventions to reduce aggressive behaviors and coercive measures (Väkiparta et al., 2019). Without such homogeneity in future research, moves towards a restraint free environment for service users will remain elusive. The continuing use of chemical restraint indicates a requirement for the education and training needs of staff in the pharmacokinetics and pharmacodynamics of the medications prescribed and the necessary ongoing medical monitoring, and this has received scant attention in the literature to date.

In EDs, restrictive interventions usually occur under a duty of care framework and associated legislation with clinical governance regarding the use of least restrictive practices (Knott et al., 2020). Further, in acute psychiatric units, service users ought to receive trauma-informed and recovery-oriented care (Hawsawi et al., 2020). Yet it seems from this research that this is not necessarily the service user experience. Indeed, a recent Australian study of current clinical practices for managing behavioral emergencies within Victorian public hospital EDs specifically acknowledges the deficit in care for the majority of service users presenting to EDs (Knott et al., 2020). Further research is needed to explore the possibilities of shared agreed interventions between nurses and service users to minimize negative experiences of chemical restraint.

The range of terms used to describe chemical restraint reinforces the continuing confusion and lack of clarity about what chemical restraint is. The interchangeability of terms such as forced medication and rapid tranquilization and reference to the use of chemical restraint when it is given as pro re nata or named as therapeutic sedation is a barrier to homogeneity in research studies (Knott et al., 2014). This points to the need for uniform understanding internationally and standardization in definition that will allow research findings to be compared more usefully than is currently the case.

6.1 | Limitations

This study was limited by the lack of published qualitative studies that focus solely on the experience of chemical restraint, outside of the context of other coercive practices. Furthermore, given the power imbalances present in mental health settings, the lack of transparency in reporting the relationship between researcher and participants across many of the studies has the potential to limit the validity of the data reporting on experiences of chemical restraint. The data were limited to studies written in English and therefore data from studies in other languages might have been missed. A further limitation relates to the inability, at times, to separate out whether the data related to chemical or other restraints. This qualitative synthesis, like all qualitative research, is furthermore limited in terms of generalizability.

7 | CONCLUSION

In the context of an increasing focus and amount of literature available about the use of chemical restraint, this paper brings together for the first time, we believe, the qualitative research specific to chemical restraint on the experiences regarding its use from the perspectives of service users and staff. The results provide important insights for clinicians and researchers about the ongoing disconnect of the nature of experiences from both parties amid the context of the risk and safety imperative of health services that promote coercive practices (albeit as a last resort). The findings also demonstrate the need for focused qualitative research on chemical restraint that is carried out separately and distinct from other coercive practices. Furthermore, there is a need for future research in this area to be from more diverse settings and to more explicitly address power imbalances.

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CONFLICT OF INTEREST

The authors declare there are no conflicts of interest.

AUTHOR CONTRIBUTIONS

Study design: Eimear Muir-Cochrane, Candice Oster. Data collection: Eimear Muir-Cochrane, Candice Oster. Data analysis: Eimear Muir-Cochrane, Candice Oster. Manuscript writing: Eimear Muir-Cochrane, Candice Oster.

AUTHORSHIP STATEMENT

EM-C and CO contributed to the conception and design of the study, collection of data, data analysis and interpretation, drafting the manuscript, and critically reviewing the manuscript. All authors are in agreement with the manuscript.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created in this study.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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